STUDENT PERFORMANCE PREDICTOR

Project Report

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Course: AIML

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This project makes predictions about how well students perform in school via machine learning. The way this works is by receiving data inputs like student's attendance, previous grades, study hours, sleep hours, as well as grades received on assignments (assignment scores). The project then uses linear regression to produce an estimate of what each student will receive as their final grade based on input data. Additionally, a graphical user interface (GUI) designed with Tkinter to be easy to use has been created for the system that makes it user-friendly. Data visualizations using graphs that show trends in student success through the analysis of multiple inputs has been included in the project.

INTRODUCTION

Machine learning can be used to analyze large data sets (such as the amount of time that students spend studying or doing homework) in order to help businesses, schools, and other organizations determine how best to improve student performance and help students succeed in school.

Problem Statement :

Many pupils do not understand how their regular routines interact with their grades and there is no easy-to-use mechanism to anticipate what grade he/she would have for any activity performed. This project will develop a prediction tool that resolves this question.

Goal

- To learn fundamental concepts of Machine Learning.
- To utilize Linear Regression.
- To develop a GUI-based application.
- To predict students' final grades.
- To provide a graphical representation of data.

This is the very important section of the report, the methodology section

Methodology - How we did it?

The methodology section is very important. Below are the steps we took to make it easy to understand :

- We created a dataset containing student information
- We loaded the data using Pandas
- We selected the features we wanted to use to predict final marks (such as number of hours studied and attendance %).
- We trained the Linear Regression model.
- The user enters information through the GUI.
- The model predicts the final marks based on the inputs.
- The final marks and graph are displayed.

Technologies that were used in the implementation:

- *Python*
- *Pandas*
- *Scikit-learn*
- *Tkinter*
- *Matplotlib*
- *Pillow*

System Design:

The system is composed of 2 main parts:

1. Machine Learning Model
2. Graphical User Interface

The prediction is done by the model, and users interact with the GUI to make predictions.

HOW THE PROJECT WAS COMPLETED ?

The project is implemented in Python. We used Pandas to read the dataset. We used Linear Regression to make predictions. We used Tkinter to create the GUI interface. We used Matplotlib to create the graphs in the GUI. The user provides input to the system, and the system outputs predicted results.

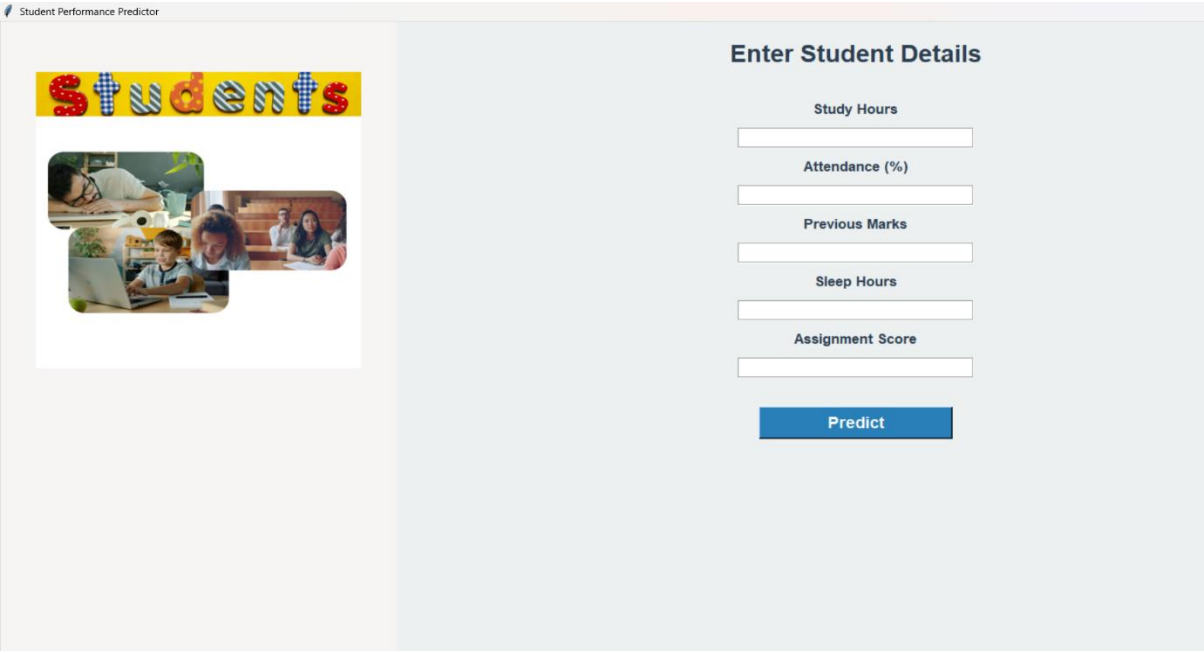
RESULTS

The system computes final marks for students based on their input. The system also generates performance categories, which can be identified as Excellent, Average or Needs Improvement and shown in the performance graph and lifestyle bar.

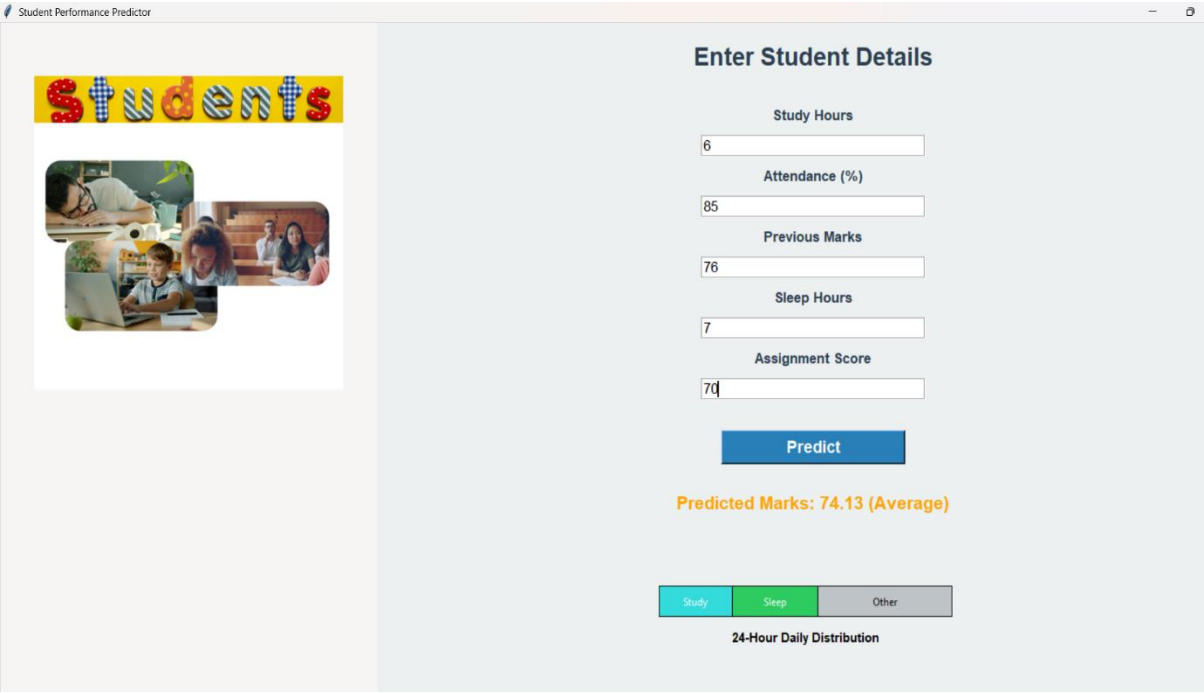
PROJECT INTERFACE:

Students Performance Predictor

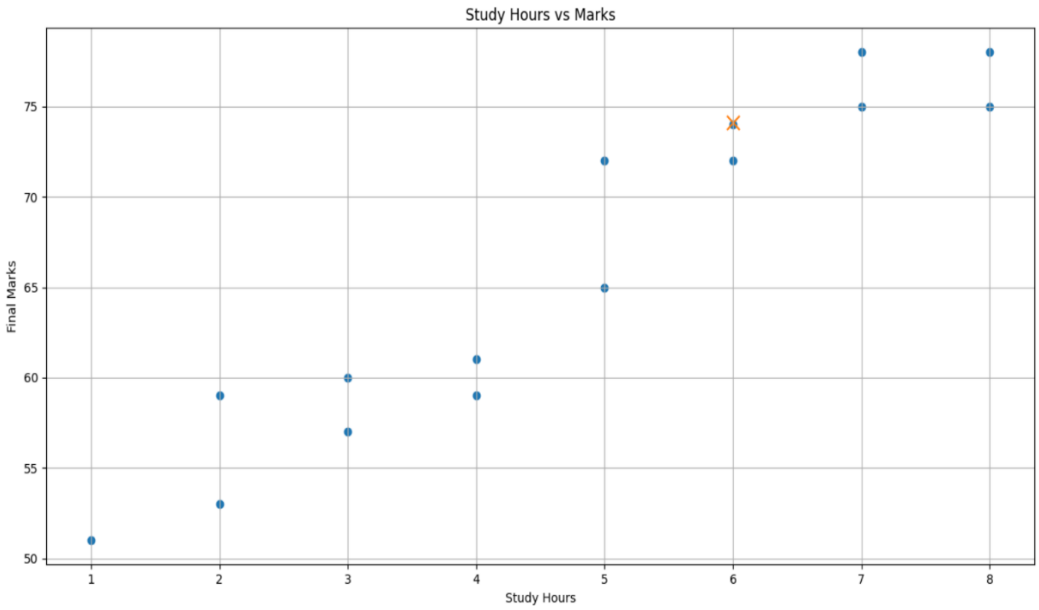
INPUT DETAILS--



OUTPUT DETAILS---



GRAPH---



DATA SET----

student_data.csv

```csv
StudyHours,Attendance,PreviousMarks,SleepHours,AssignmentScore,FinalMarks
1,50,40,5,42,51
2,60,45,6,49,59
3,65,50,6,53,60
4,70,55,7,59,61
5,75,60,7,66,72
6,80,65,7,71,74
7,85,70,8,76,78
8,90,75,8,83,75
2,55,42,5,47,53
3,60,49,6,51,57
4,68,52,6,56,59
5,72,58,7,63,65
6,78,63,7,69,72
7,82,68,8,74,75
8,88,74,8,82,78
```

BENEFITS OF THIS PROJECT:

- Ease of Use
- Rapid Prediction
- Intuitive User Interface
- Value for Students

LIMITATIONS:

- Data from data set accuracy

- Simplicity of model

- Limited to a few features

WORKS CITED:

1. Python Documentation
2. Scikit-Learn Documentation
3. Various Web-Based Tutorials

FUTURE OPPORTUNITIES :

Future enhancement for this project—

- Use a more sophisticated machine learning model.
- Include additional input parameters.
- Develop a web and/or mobile application.
- Use actual data from the students in real-time.

CONCLUSION :

Machine learning has been applied to the predict outcome of students and the combination of the machine learning with the graphical user interface (GUI) provides added value through interactivity of the total system.